

4.1 Introduction to Client Scripts

Phase 4 - Client Script Implementation | Implement Client Script & UI Policy (Incident)

1. What is a Client Script?

A Client Script is a piece of JavaScript stored on the ServiceNow server but executed on the client, that is, inside the user's web browser. It allows the form to respond immediately to events such as loading, field changes and submission, which makes the user experience interactive and responsive.

2. Types of Client Scripts

Type	When it runs	Typical use
onLoad	When the form is loaded	Set defaults, show a message on open
onChange	When a specified field value changes	React to field changes, show guidance
onSubmit	When the form is saved or submitted	Final validation before saving
onCellEdit	When a list cell is edited	Validate list edits

3. Client-side APIs

- **g_form** - get and set field values, show messages, set mandatory, visible or read-only.
- **g_user** - information about the logged-in user such as roles.
- **g_scratchpad** - passes data from server to client at form load.

4. Standard onChange Parameters

Parameter	Description
control	The form field element that changed
oldValue	Value of the field before the change
newValue	Value of the field after the change
isLoading	True while the form is loading, false for user changes
isTemplate	True when the change comes from applying a template

5. Advantages and Limitations

- Advantages: instant feedback, reduced server load and better user experience.
- Limitations: runs only in the browser UI, and it is not applied to server-side or import updates.
- Best practice: avoid heavy synchronous server calls such as GlideRecord on the client.